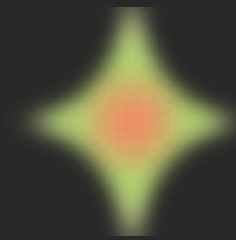
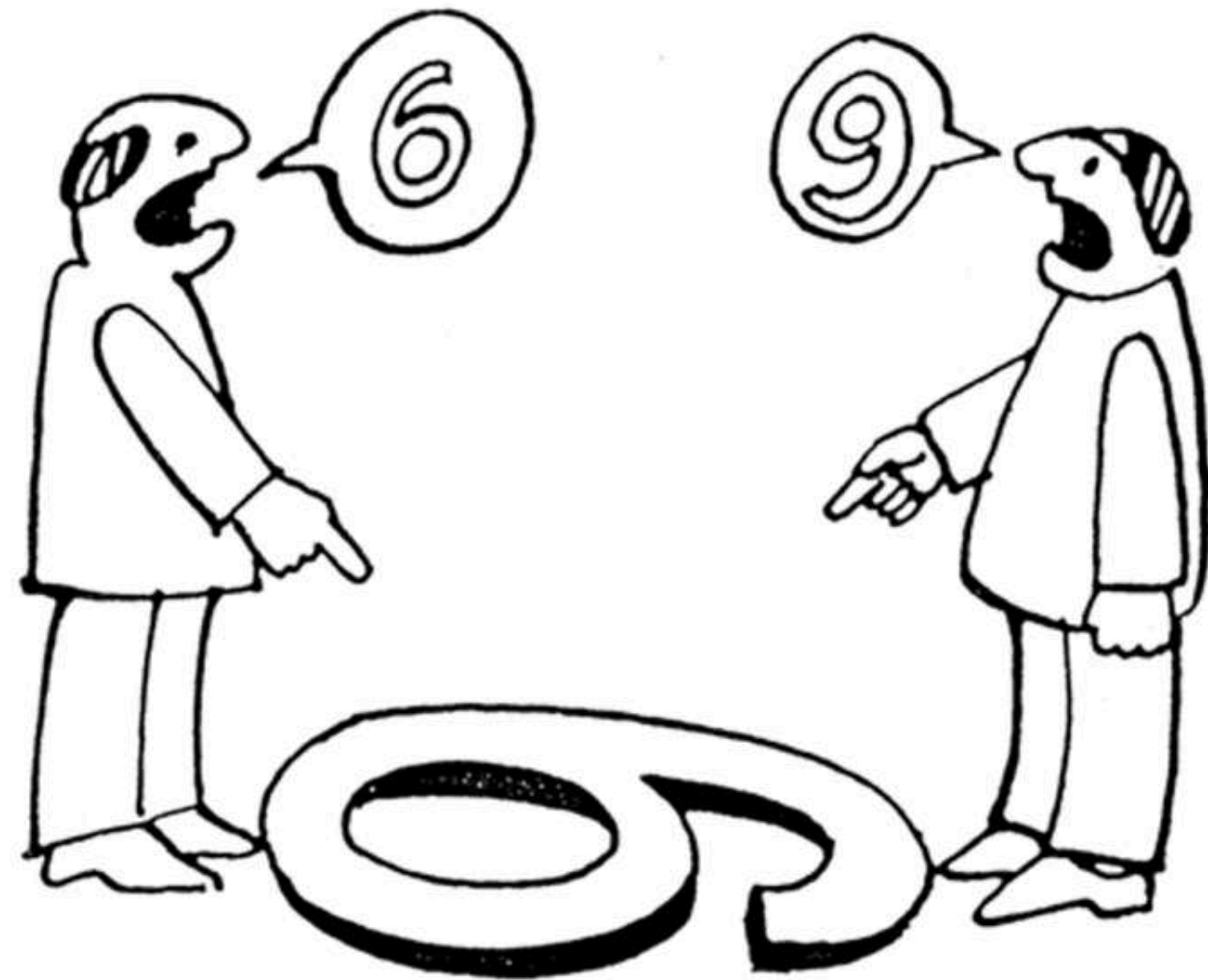


Introduction to
SYSTEMS BIOLOGY
&
BIOLOGICAL NETWORKS

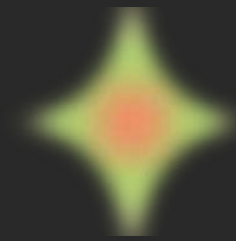
By the end of this internship, you will be able to...

1. Understand the fundamental concepts of systems biology, biological networks, and computer-aided drug discovery (CADD).
2. Interpret biological networks by identifying key topological features such as hubs, bottlenecks, modules, and network motifs.
3. Evaluate potential therapeutic targets using systems biology principles and network-based analyses.
4. Construct and analyze biological networks using computational tools.
5. Perform key CADD workflows, including protein visualization, ligand preparation, molecular docking, and virtual screening

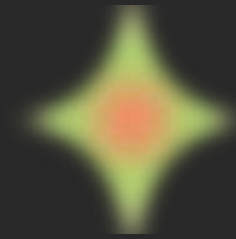
The reductionist problem



Each biologist specialises in one area of biology



Neither of them are wrong, but neither of them tell the whole story.



Biology alone isn't enough. We need physics, math, chemistry, & computer science

What could they have done to avoid the disagreement?

- ✦ **Communicate and explain their perspectives:** Instead of arguing about who is right, they could ask, "What do you see?" and understand why the other person reached a different conclusion.
- ✦ **Gather more context:** Is there any indication of which side is "up"? Is it part of a larger sign or document? Additional information can resolve the ambiguity.
- ✦ **Consult an independent source of information:** Ask the person who wrote the number, check the intended orientation, or look for another reliable reference.
- ✦ **Accept that one observation may not be sufficient:** Before drawing a conclusion, recognize that a single viewpoint can be incomplete or ambiguous.

Systems biology helps us see the big picture

- ✦ Integrating perspectives
- ✦ Changing viewpoints
- ✦ Gathering more context
- ✦ Consulting independent evidence
- ✦ Combining observations



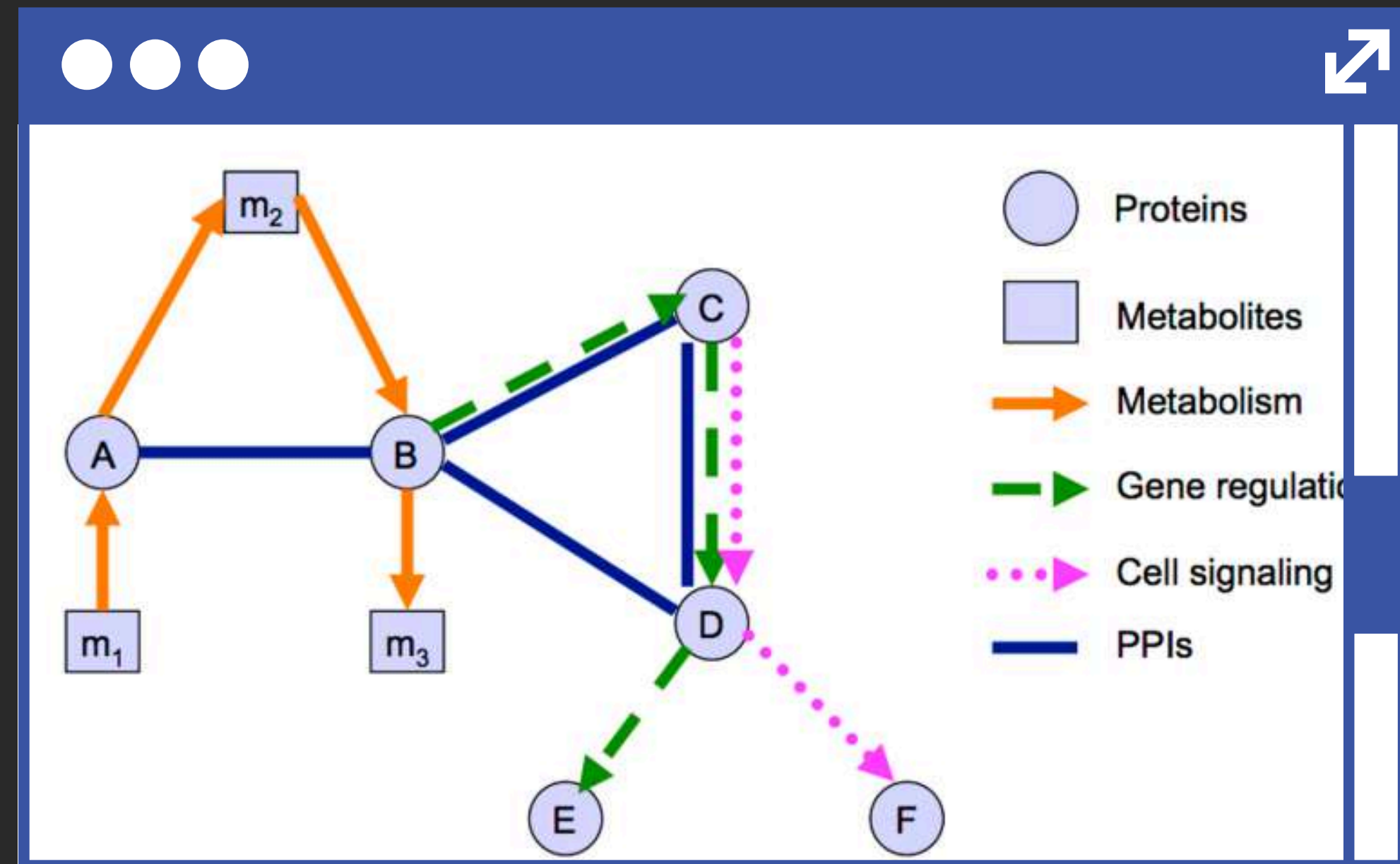
Systems biology

- ✦ Systems biology is a way of thinking before it is a set of tools. It is an interdisciplinary field that uses computational and mathematical models to analyze the complex interactions within biological systems.
- ✦ Rather than studying isolated components, it takes a holistic approach to understand how genes, proteins, and cells function together as an integrated network.
- ✦ Systems biology aims to understand biological entities at the systemic level, analysing them not only as individual components, but also as interacting systems and their emergent properties.

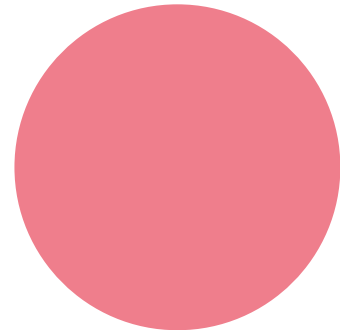
**Partial observation → Different interpretations → Conflict → Reflection
→ Integration → Complete understanding**

Biological networks & graph theory

- ✦ It is the mathematical modeling of complex biological systems as networks. It translates intricate biological data into structures composed of nodes (vertices), which represent biological entities (genes, proteins, cells, or species), and edges (arrows), which represent their interactions or relationships (regulation, binding, or predation)
- ✦ It deals with the properties of graphs
- ✦ Network biology allows the representation and analysis of biological systems using tools derived from graph theory.



Graph theory – Properties of graphs



Node/Vertex

- Represents an entity: a protein, a gene, a metabolite, a cell type
- Two nodes connected to each other by one edge are called neighbours



Edge

- Represent interactions, relationships, or connections between nodes (e.g., protein-protein interactions, gene regulation, metabolic reactions)



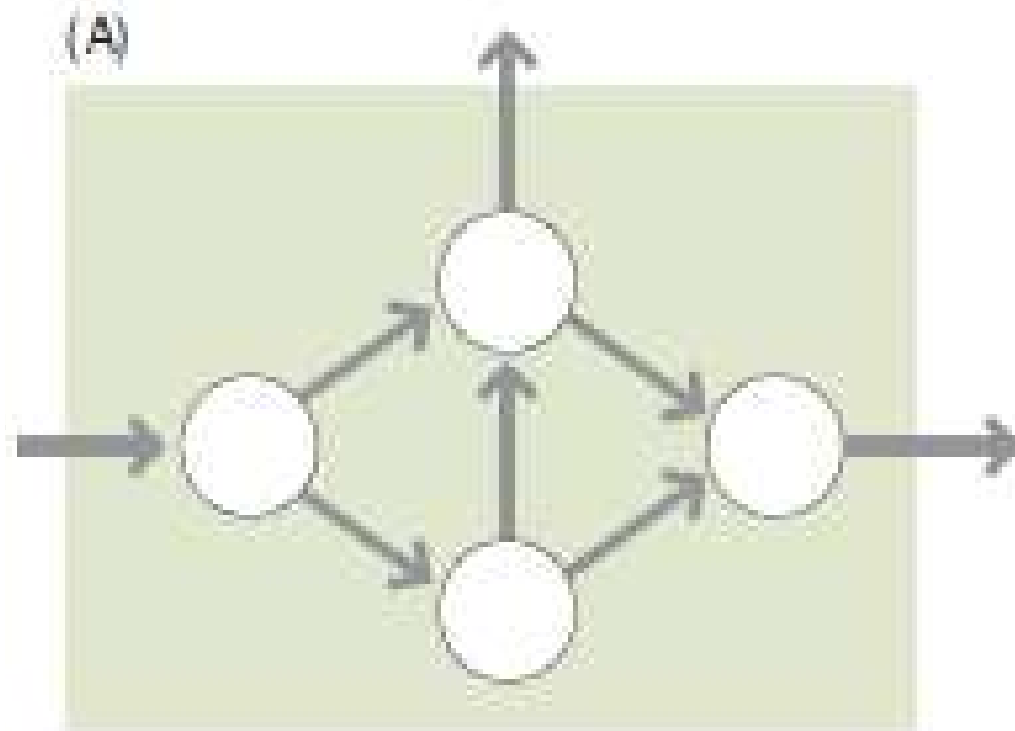
Repression



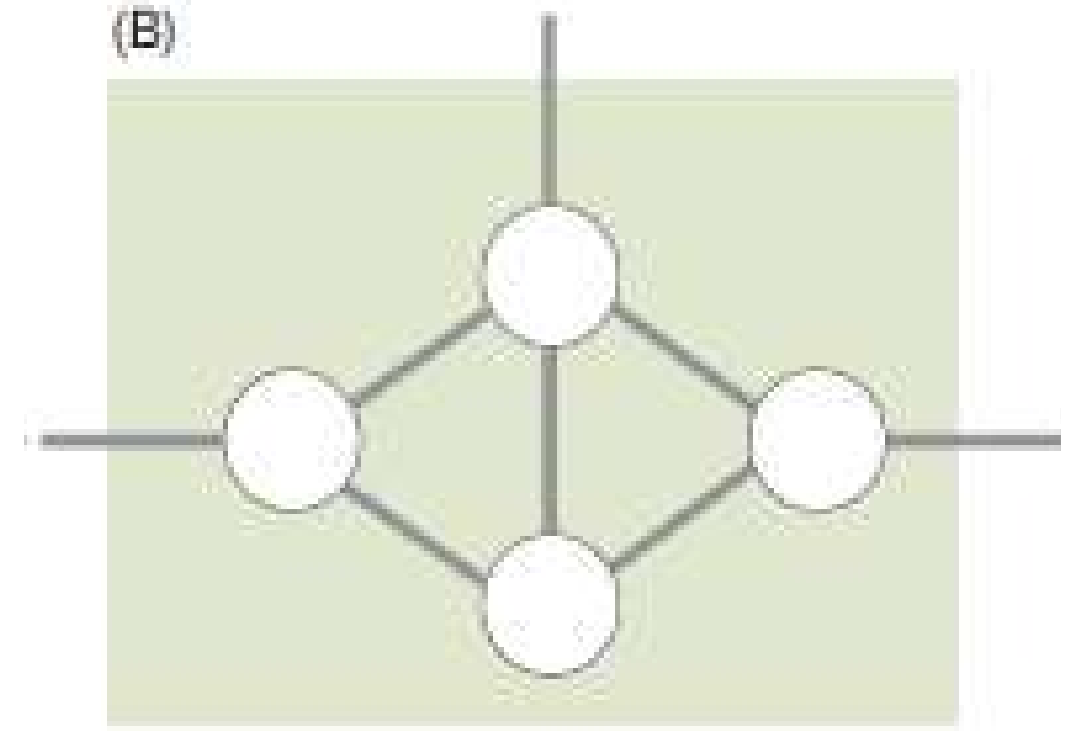
Activation

Types of edges

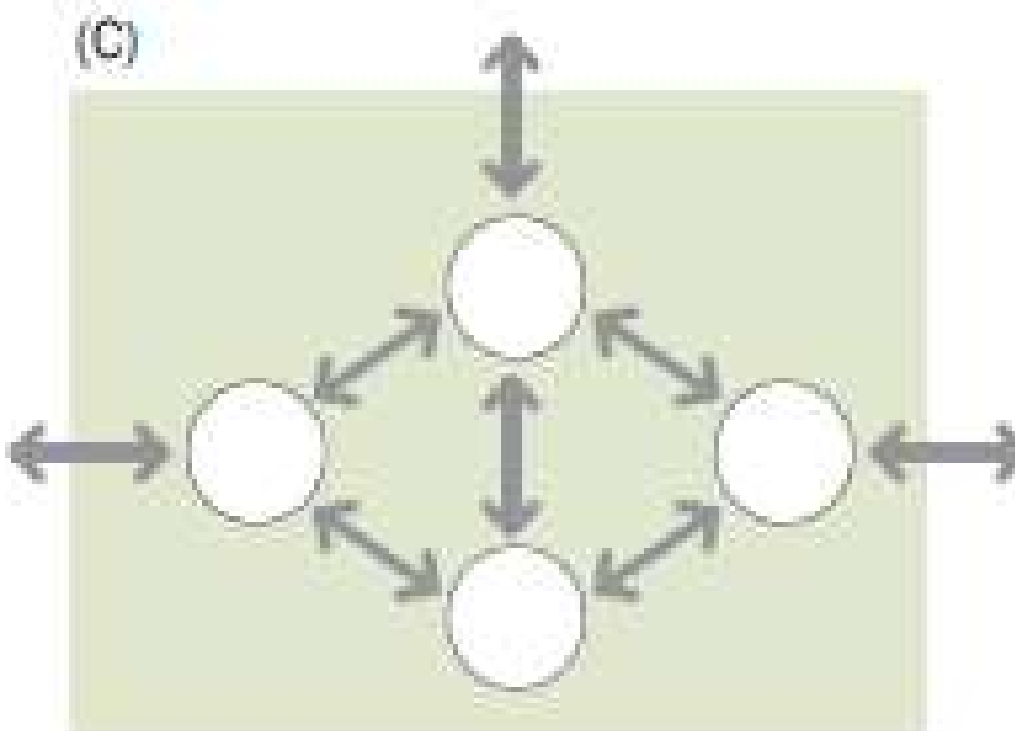
Directed



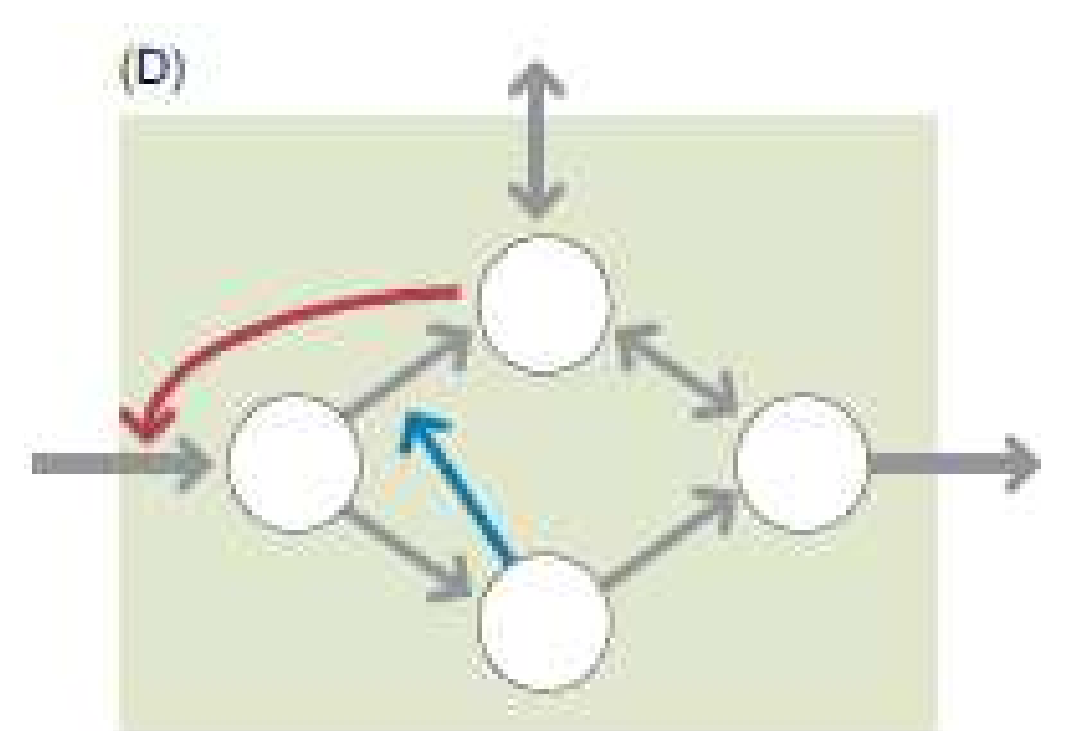
Undirected



**Bidirectional
edges**



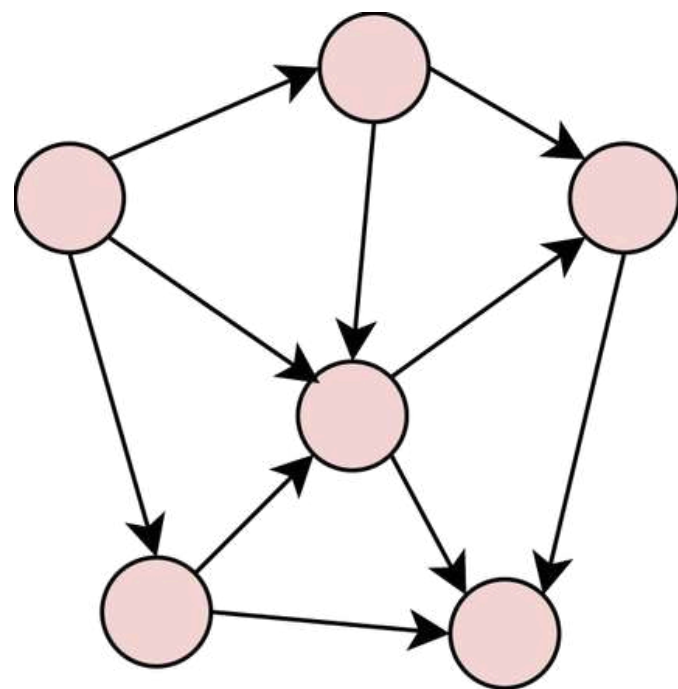
**Regulation
arrows**



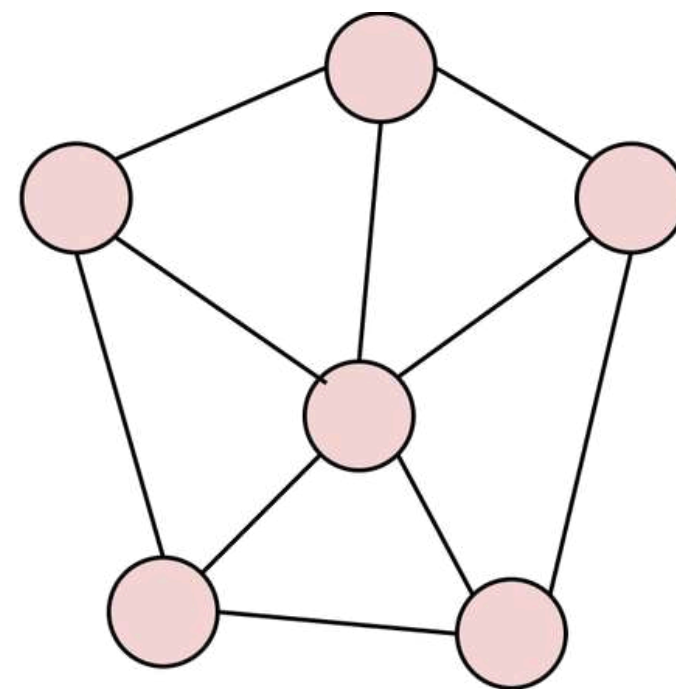
Basic network components

Graphs can be:

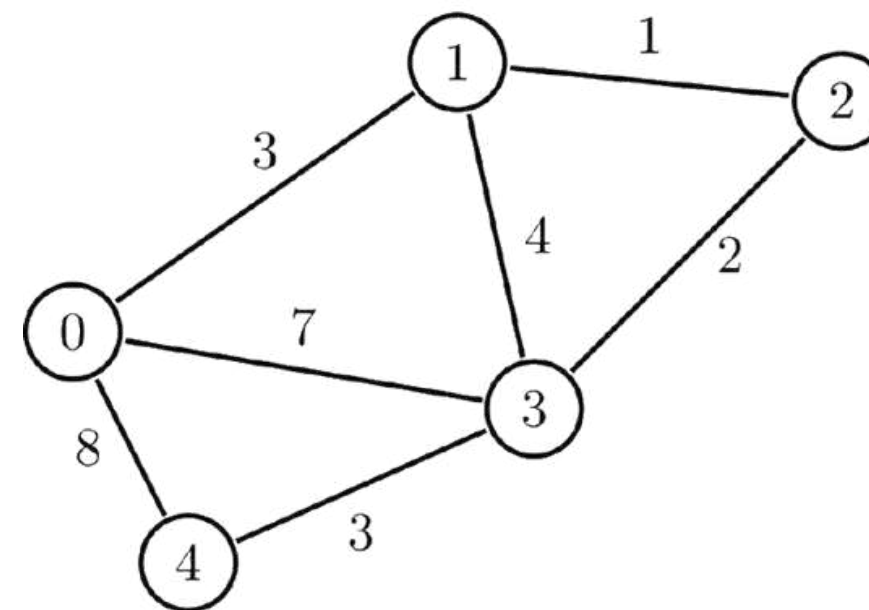
- **Directed:** Edges have direction (e.g., gene regulatory networks, signaling pathways)
- **Undirected:** Edges have no direction (e.g., protein-protein interactions)
- **Weighted:** Edges have weights representing interaction strength or confidence
- **Bipartite:** Two distinct types of nodes (e.g., transcription factors and their target genes)



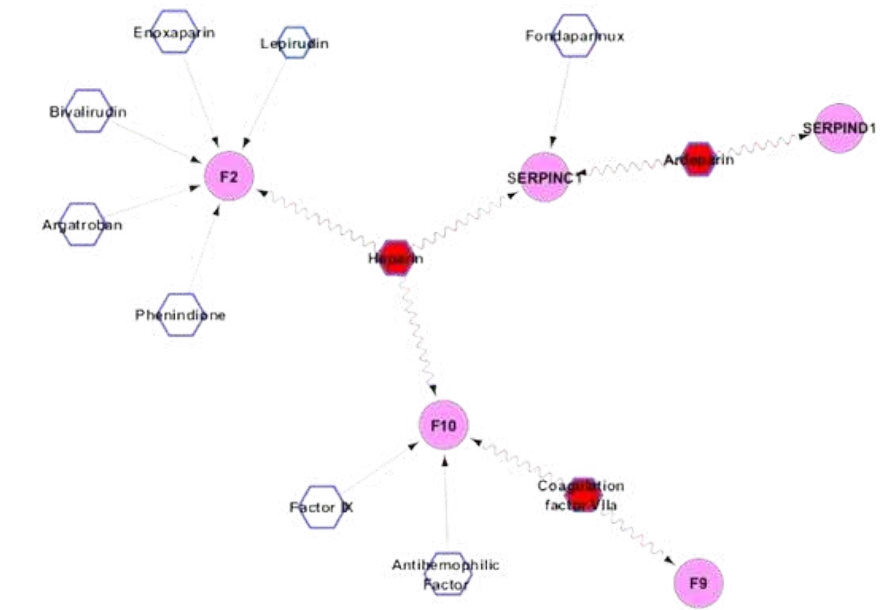
Directed Graph



Undirected Graph



Weighted



Bipartite

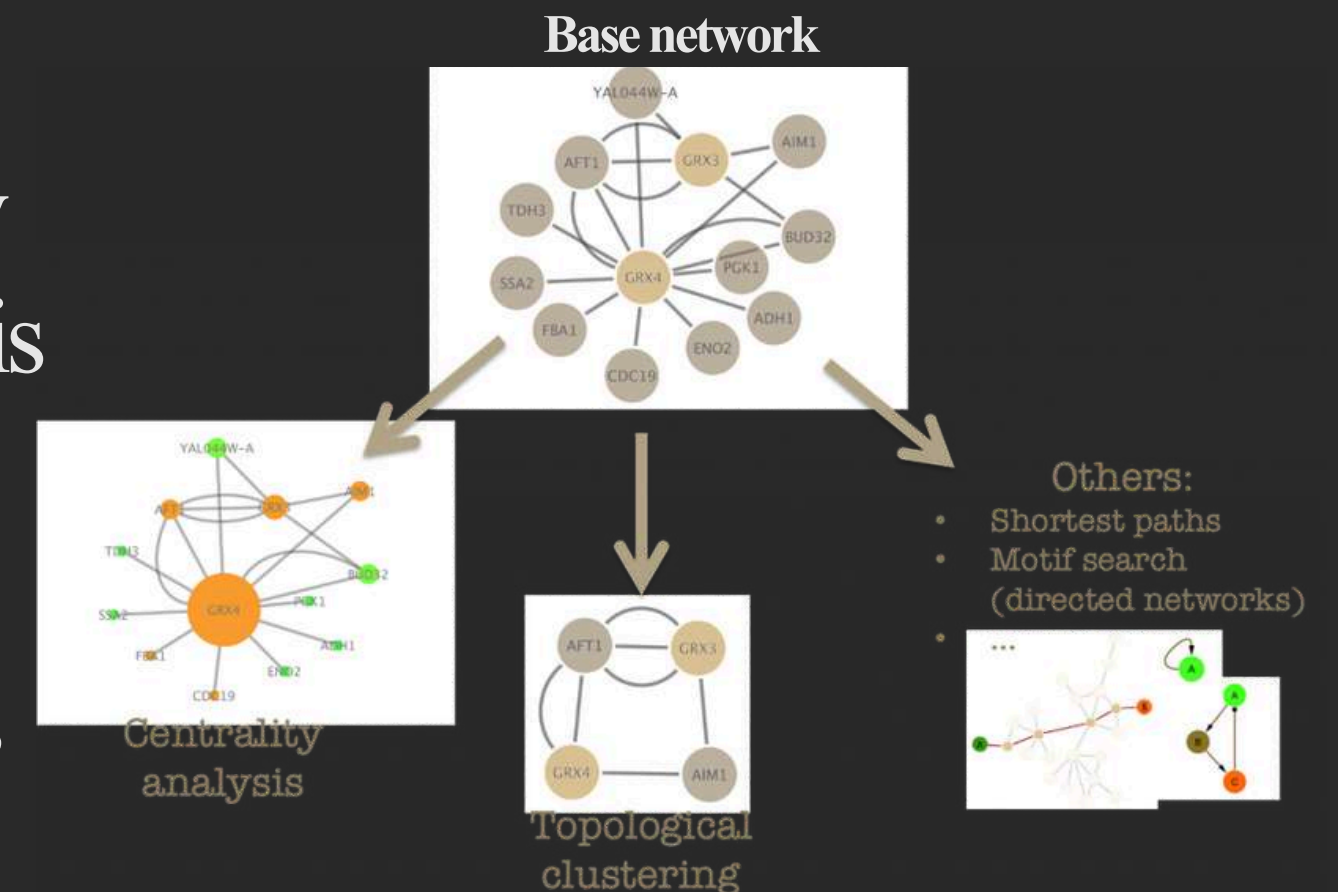
Network biology

✦ **Topology** is the way in which the nodes and edges are arranged within a network.

✦ Topological properties can apply to the network as a whole or to individual nodes and edges.

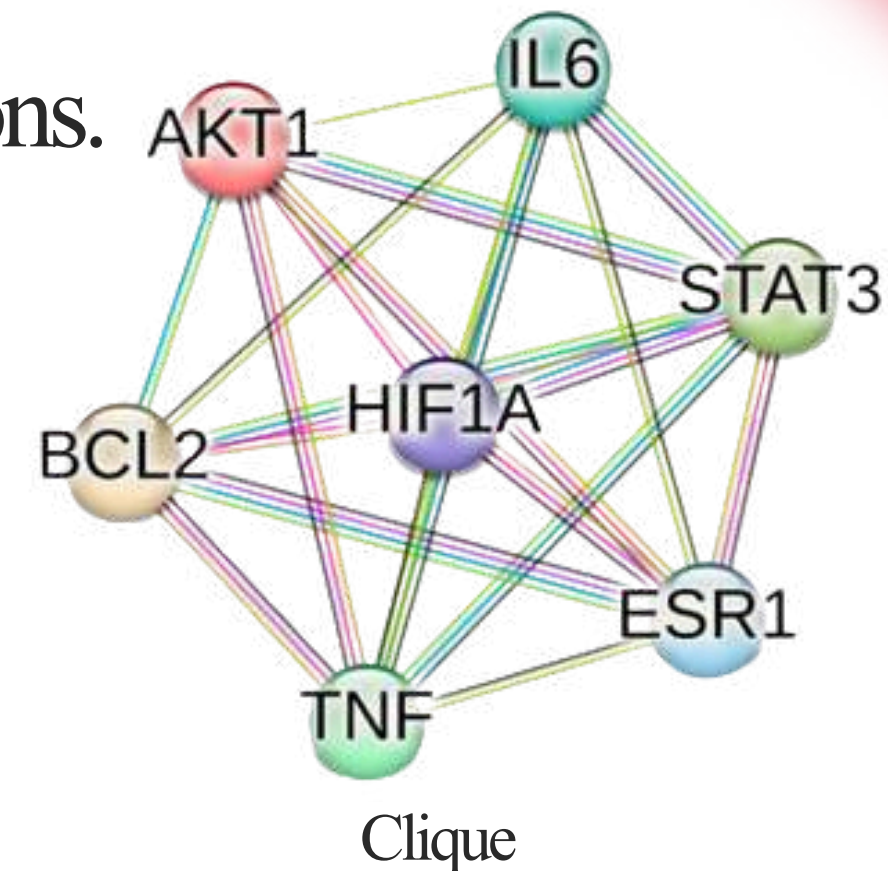
✦ Networks have a number of characteristics, mainly:

- **Small-world effect:** Network diameter is usually small (~ 6 steps), no matter how big the network is
- **Scale-free:** A small number of nodes (hubs) are lot more connected than the average node
- **Transitivity:** The networks contain communities of nodes that are more connected internally than they are to the rest of the network



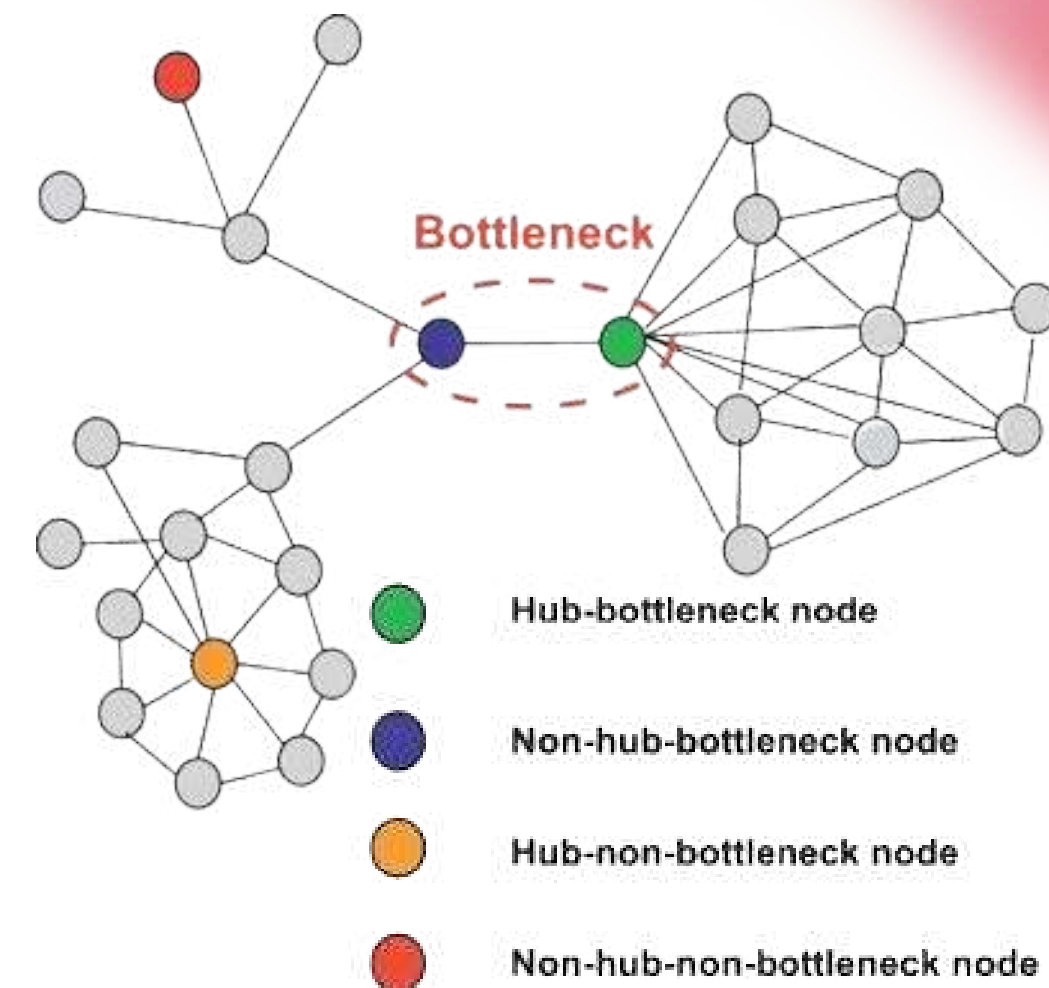
Network topology

- The **degree (k)** of a node is the number of edges connected to it. In directed networks, nodes have in-degree (incoming edges) and out-degree (outgoing edges).
- The **degree distribution $P(k)$** describes the probability that a randomly selected node has degree k .
- The **clustering coefficient (C)** measures whether the neighbours of a given node will form a clique, i.e. the tendency of nodes to form triangles or clusters. It is the ratio of actual connections between its neighbors to the maximum possible connections.
- A **clique** is a fully connected graph
- Network density is the ratio of actual edges to the maximum possible edges in a network.



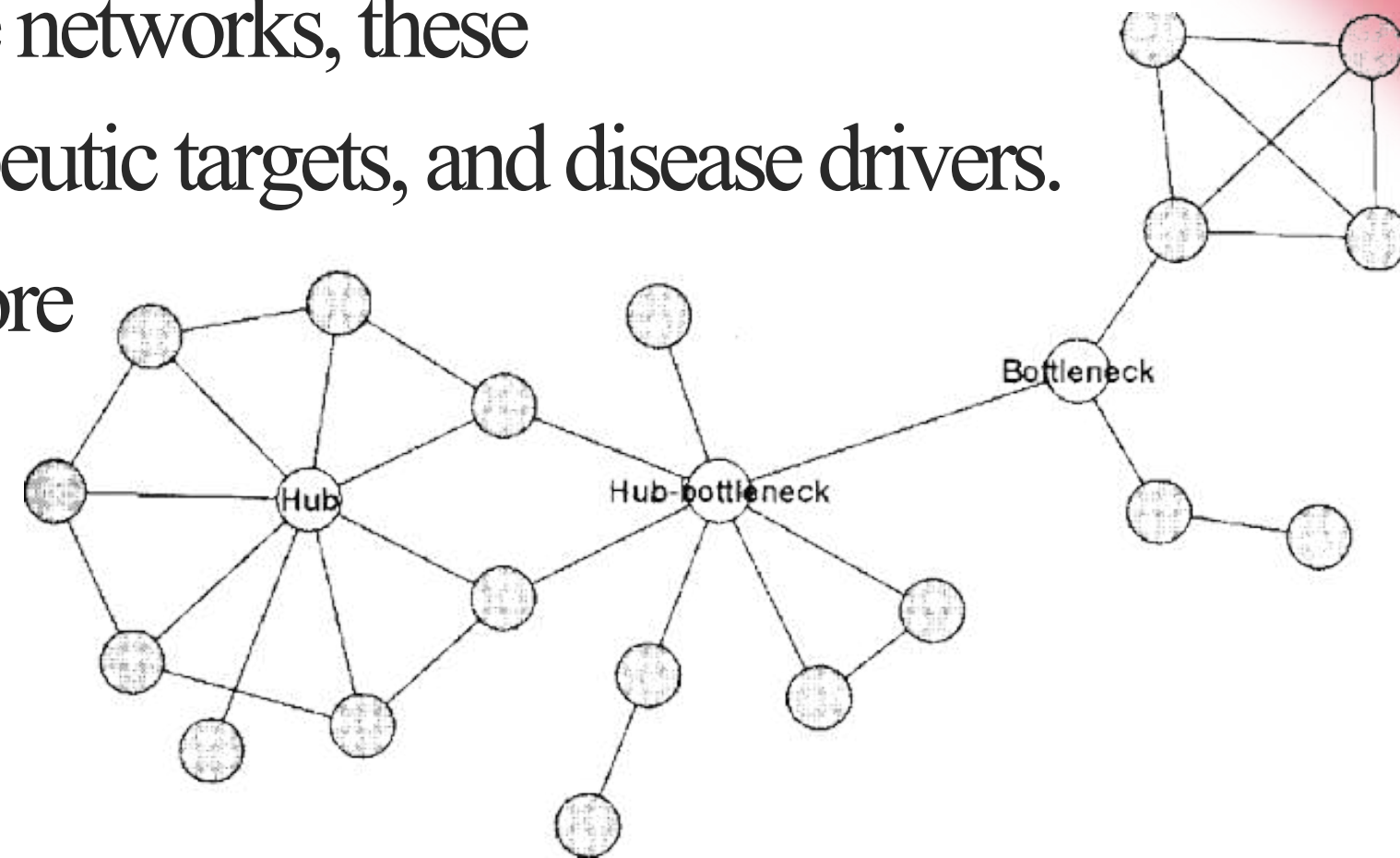
Centrality measures

- Centrality metrics quantify the relative importance or influence of nodes in a network.
- **Degree centrality** equals the number of connections a node has. Nodes with high degree centrality (called **hubs**) are often essential genes or proteins critical for network function. However, this is a local measure since it does not take into account the rest of the network and the importance we give to its value depends strongly on the network's size.
- **Betweenness centrality** measures how often a node lies on the shortest paths between other node pairs. Nodes with high betweenness act as **bottlenecks** controlling information flow and are often essential for network integrity.
- **Closeness centrality** measures how close a node is to all other nodes, defined as the inverse of the sum of shortest path distances



Hubs & bottlenecks

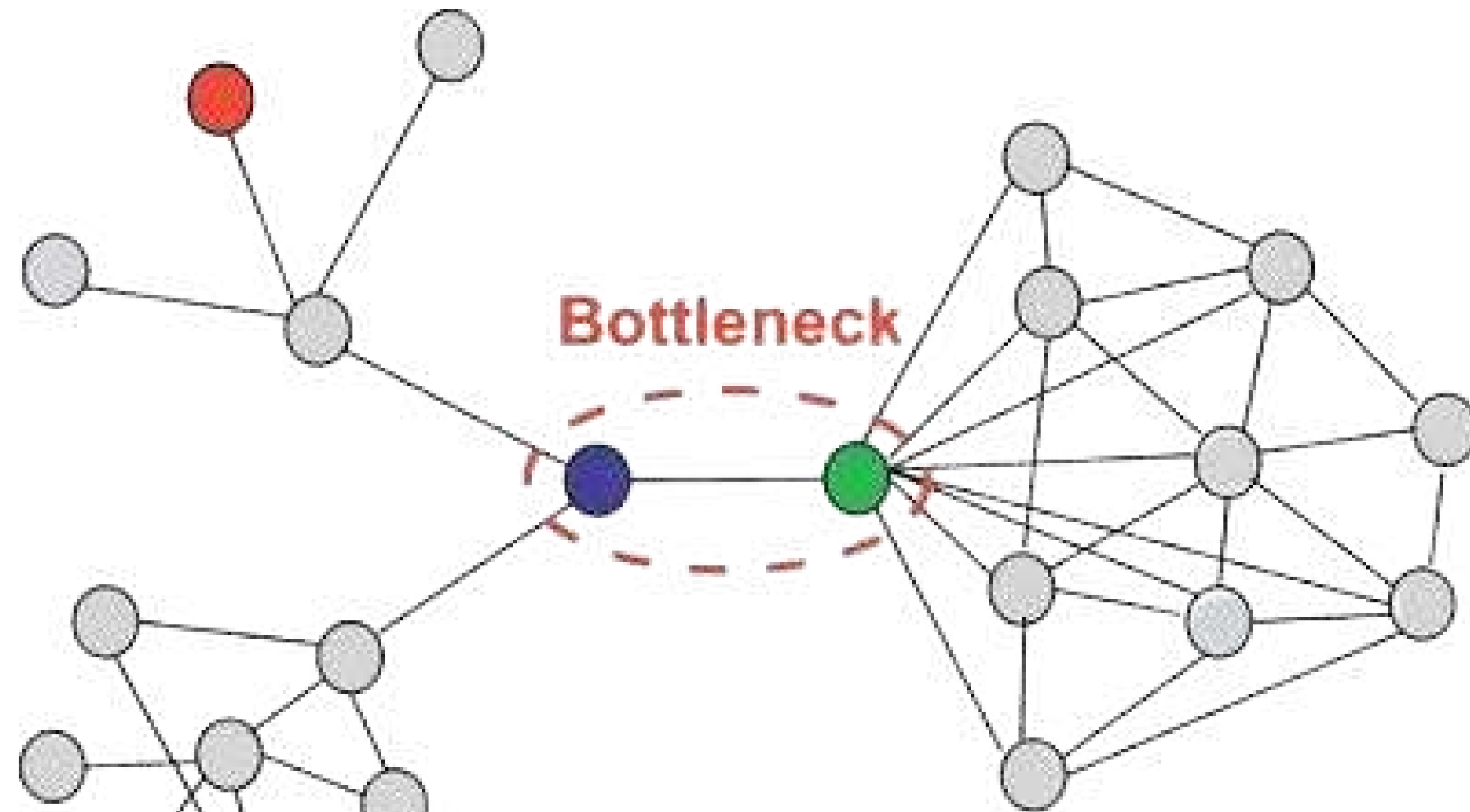
- **Hub nodes:** High degree, i.e. connected to many other proteins. Removing a hub disrupts local connectivity.
- **Bottleneck nodes:** High betweenness, i.e. sits on many shortest paths between other proteins. Not necessarily a big junction, but removing it isolates whole regions.
- **Hub-Bottleneck nodes:** The most biologically important. A node that satisfies both criteria (many connections AND a bridge position). In disease networks, these proteins are the most likely to be essential genes, therapeutic targets, and disease drivers.
- **Modules (communities):** groups of nodes that are more connected internally than to the rest of the network. Often correspond to functional units (a signaling complex, a metabolic pathway branch, a co-regulated gene set...)



Ganging up!

- **Community / Cluster** : A group of nodes that are more connected within themselves than with the rest of the network. In networks, communities fall into two categories: **functional modules and protein complexes**.
- **Module**: Modules are exchangeable functional units in which the nodes (proteins) do not have to be interacting in the same time or space. An important characteristic of a module is that its intrinsic functional properties do not change when it is placed in a different context.
- **Complex**: A group of proteins that interact with each other at the same time and in the same space, forming relatively stable multi-protein machinery.
- **Motif**: Statistically over-represented sub-graphs in a network. They correspond with a pattern of connections that generates a characteristic dynamical response (e.g. a negative feedback loop).

Hubs, bottlenecks, & modules



Hub-bottleneck node



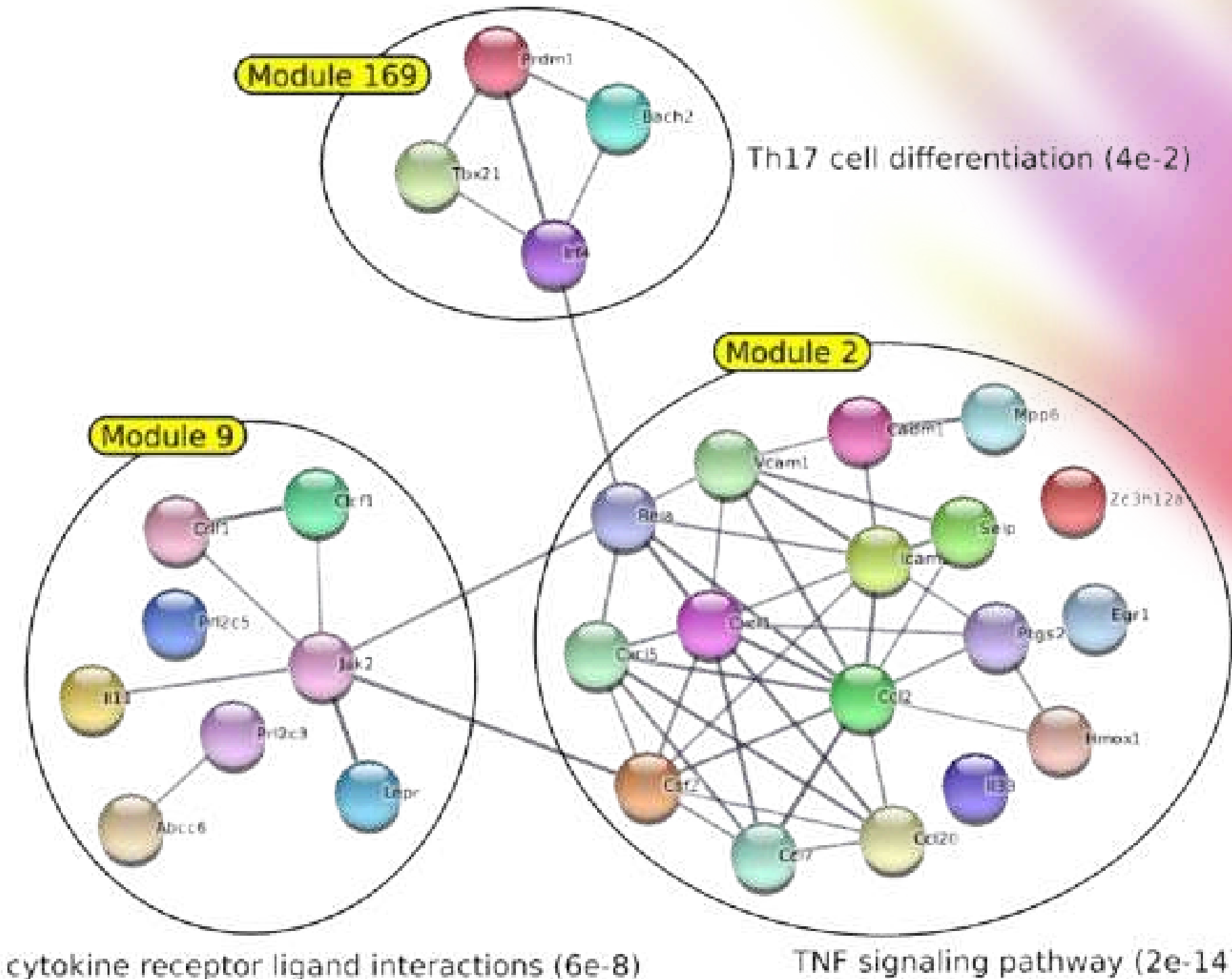
Non-hub-bottleneck node



Hub-non-bottleneck node



Non-hub-non-bottleneck node



IL-6-type cytokine receptor ligand interactions (6e-8)

TNF signaling pathway (2e-14)

- Metabolic network

Types of biological networks

- Protein-protein interaction

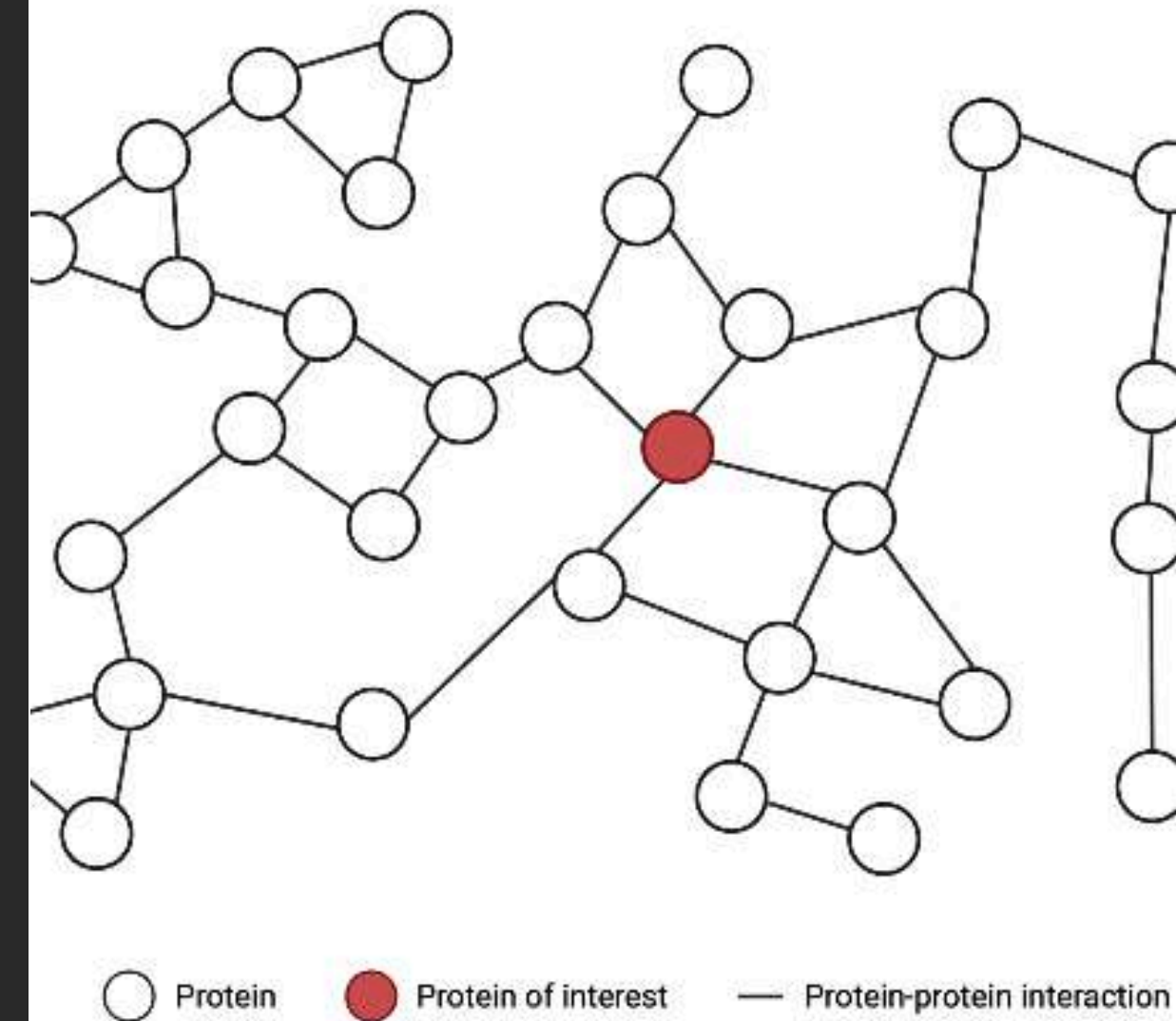
- Gene regulatory network

- Genetic interaction

- Signalling network

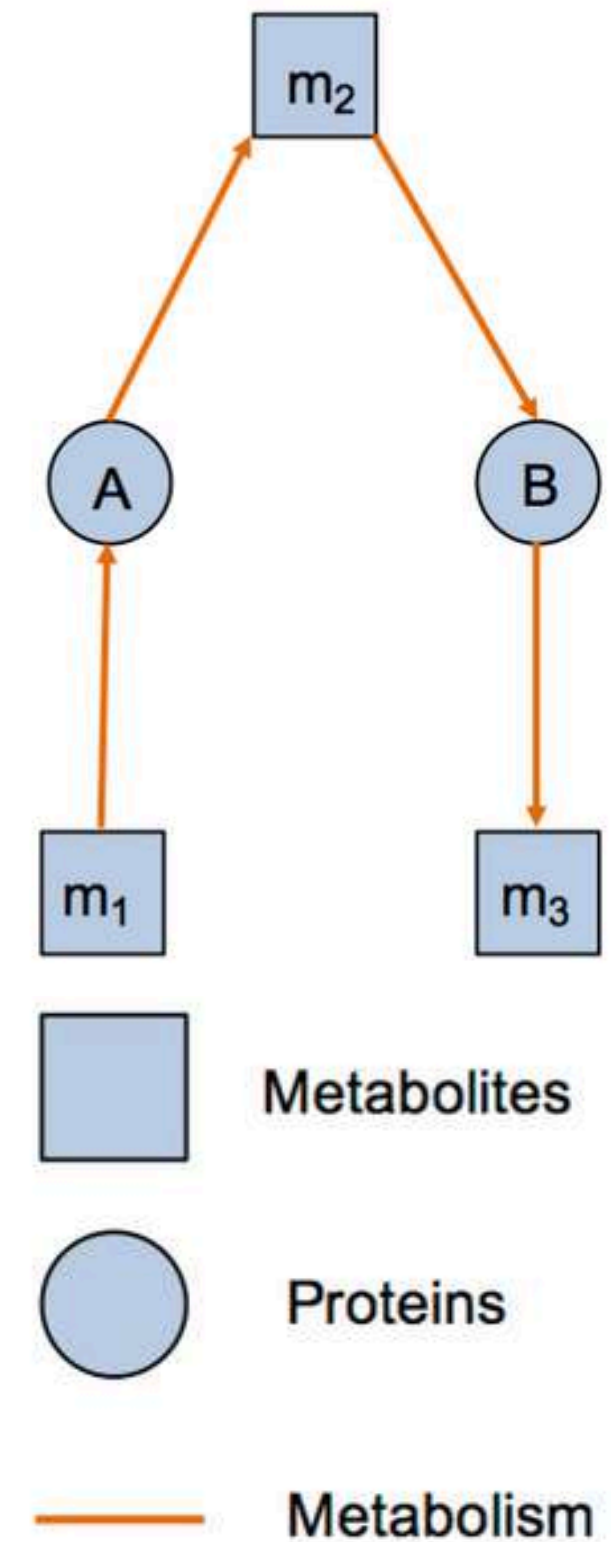
PPI networks

- It is a visual and mathematical map of the physical or functional contacts between proteins in a cell.
- These contacts: 1] are specific, 2] occur between defined binding regions in the proteins, 3] have a particular biological meaning (i.e., they serve a specific function)
- Most PPI networks are undirected: "A binds B" means the same as "B binds A.", whereas Signaling networks are directed: "A phosphorylates B" is not the same as "B phosphorylates A."



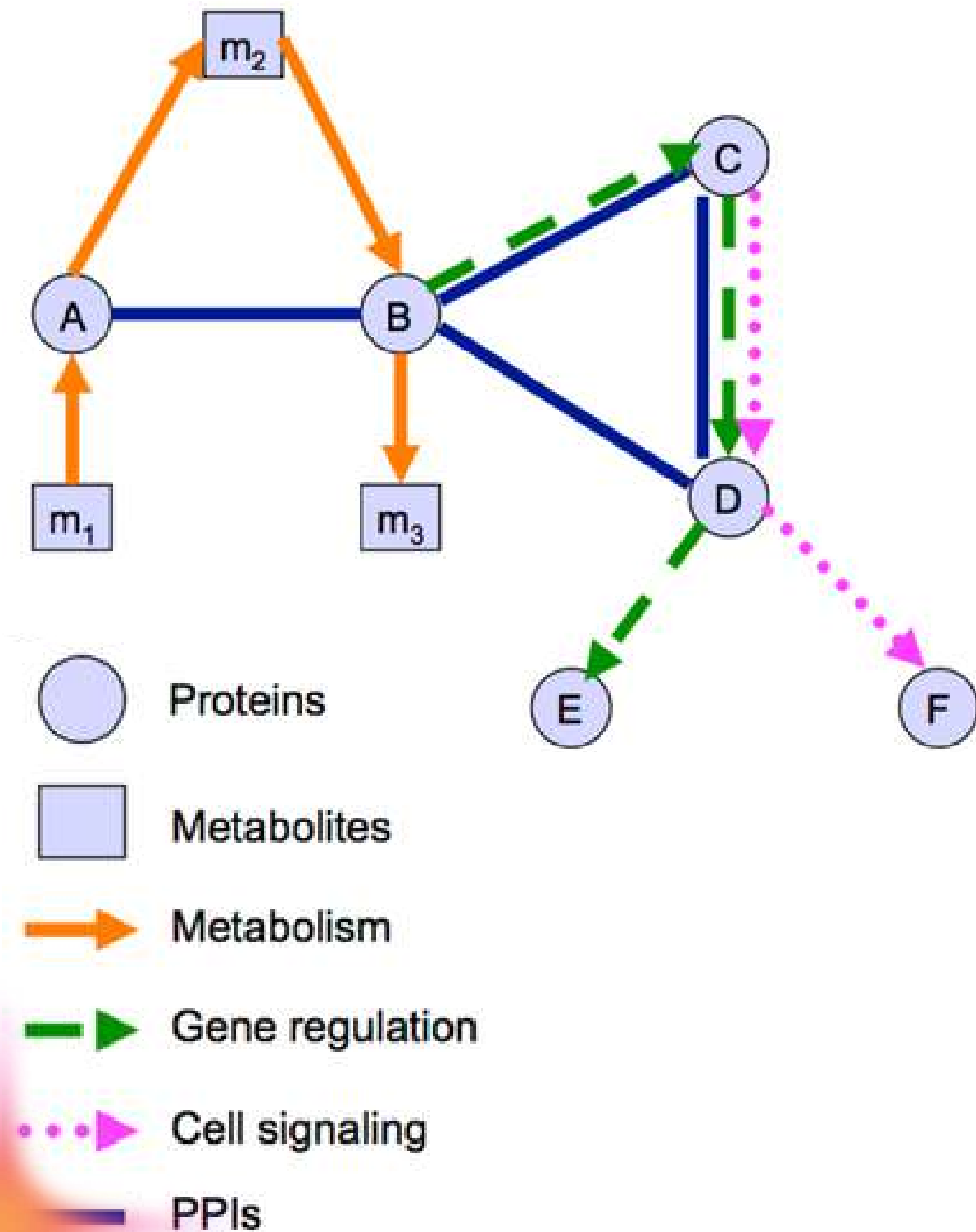
Metabolic networks

- Represent the biochemical reactions that allow an organism to grow, reproduce, respond to the environment and maintain its structure.
- Metabolites and enzymes take the role of nodes and the reactions describing their transformations are represented as directed edges
- Edges can represent the direction of the metabolic flow or regulatory effects of a specific reaction



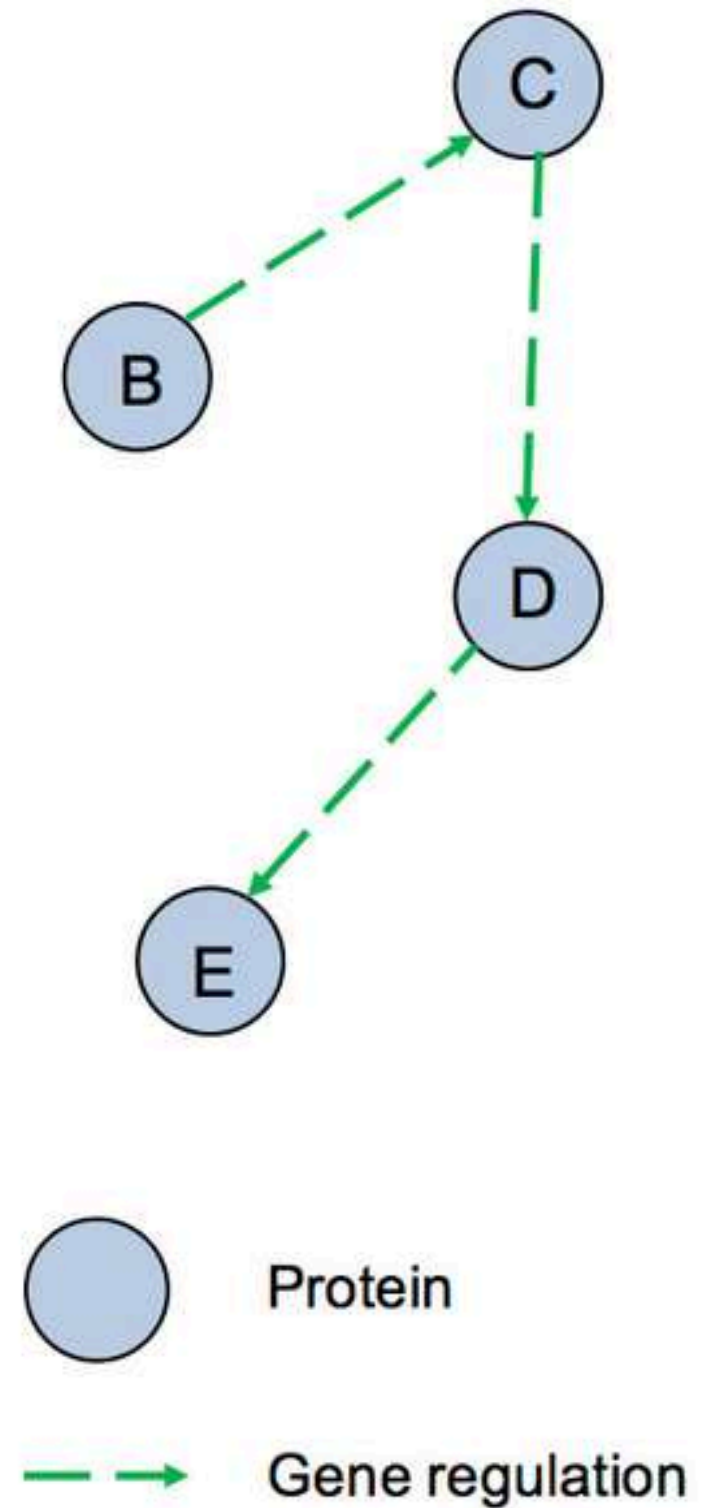
Genetic interaction network

- Measures the unexpected phenotypes (such as fitness) that occur when multiple genes are mutated simultaneously. Represent a functional relationship between different genes, rather than a physical one
- Genes are represented as nodes and their relationships as edges.
- Depending on the type of evidence behind their interaction, directionality can be inferred for the edges



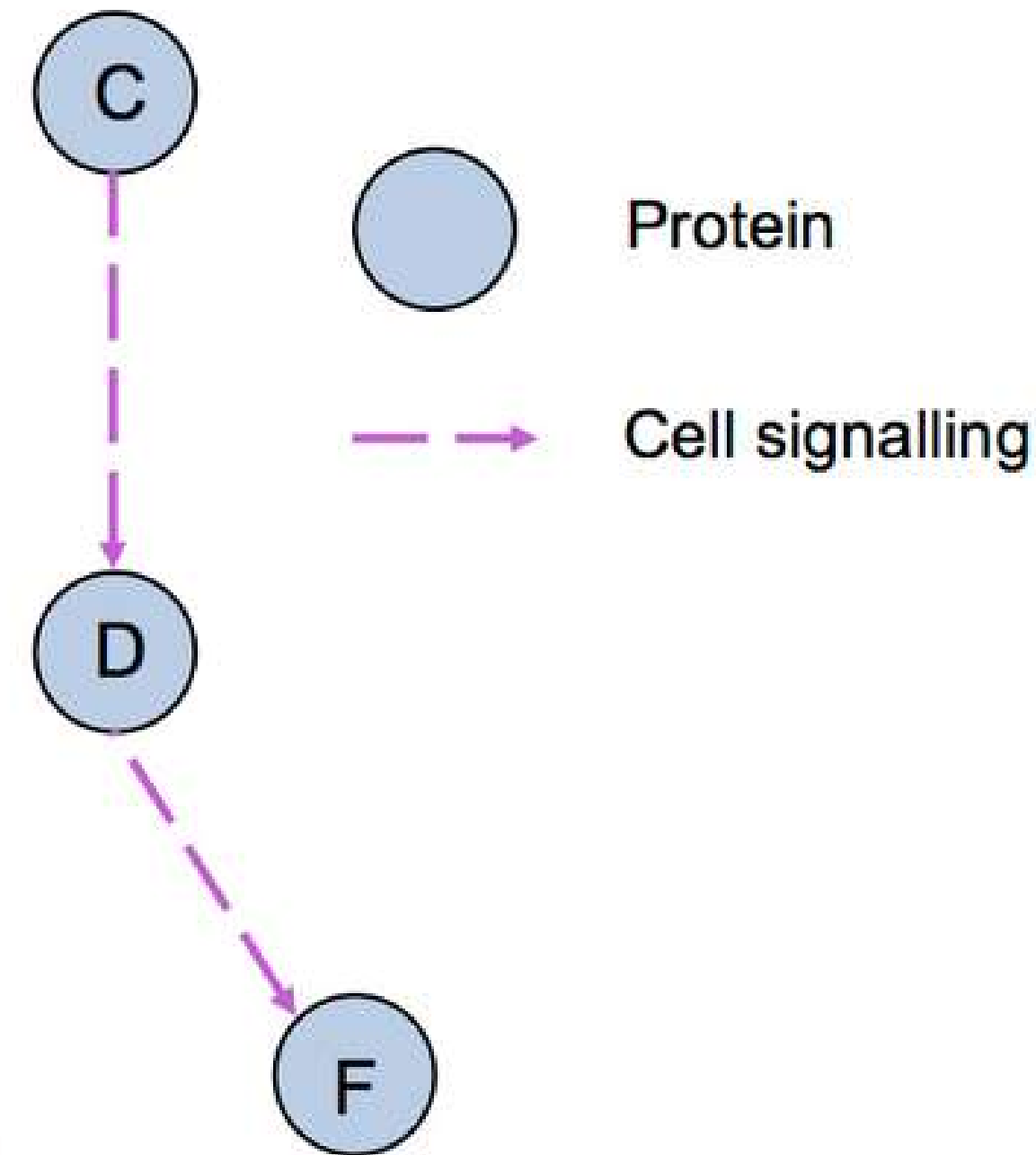
Gene regulatory network

- Represent how gene expression is controlled.
- Genes and transcription factors are represented as nodes, while the relationship between them is depicted by different types of directional edges.
- Regulatory RNAs and other mechanisms can also form part of this type of network
- Usually generated via databases representing consensus knowledge of gene regulation (e.g. Reactome, KEGG)

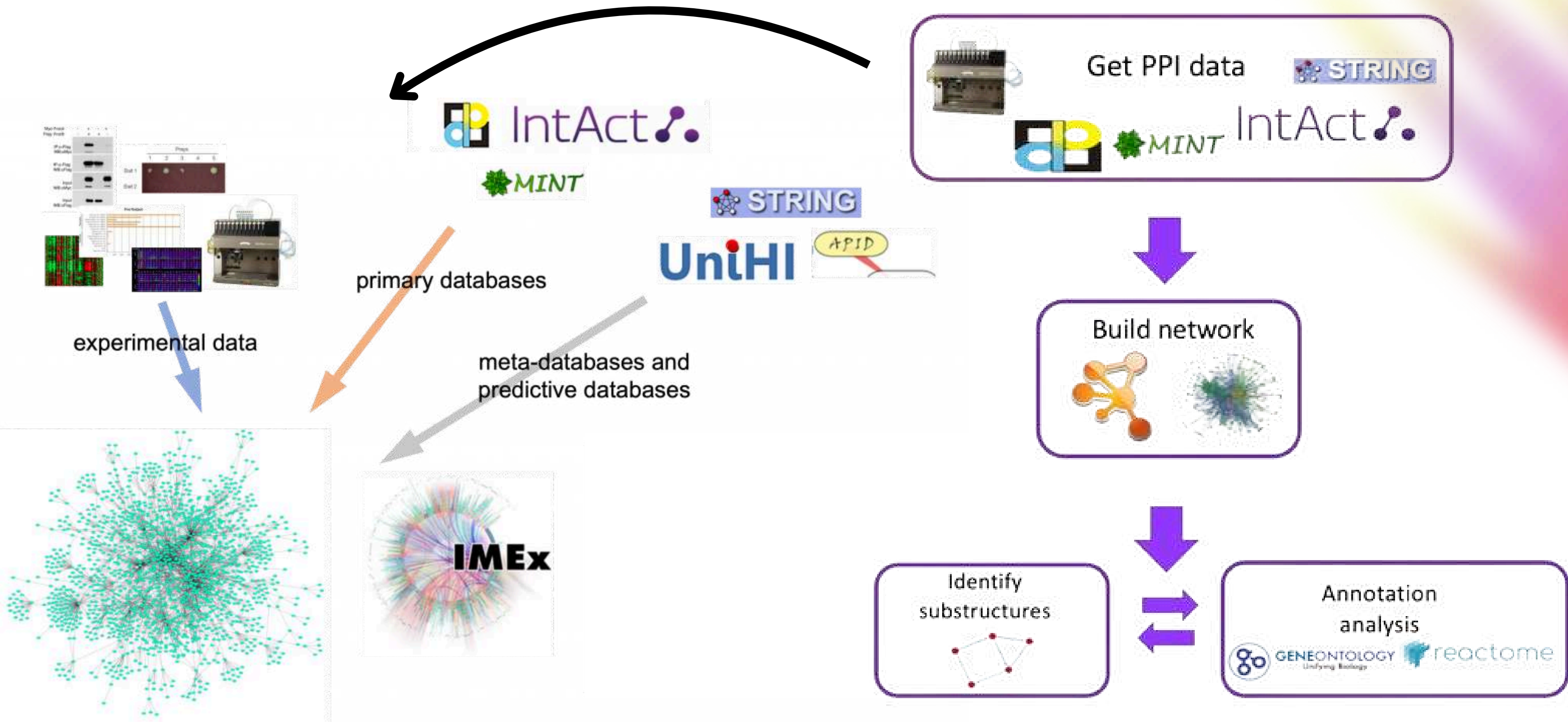


Signalling network

- Cell signalling is the communication system that controls cellular activities.
- Signalling pathways represent the ordered sequences of events and model the information flow within the cell.
- Gene regulation networks can be considered as a sub-type of cell signalling networks, focusing on a specific signalling event which is often the final stage of a signalling cascade.
- Elements in the pathway (e.g. proteins, nucleic acids, metabolites) are represented as nodes and the flow of information is represented by directed edges.
- They are systematically represented by two types of resources: 1] Pathway databases (e.g. Reactome, KEGG) and 2] Reaction network databases/Activity flow resources (e.g. Signor, SPIKE)



Systems biology workflow - From disease to network



LEARN HOW TO SEE. REALIZE THAT
EVERYTHING IS CONNECTED TO
EVERYTHING ELSE.



- *Leonardo Da Vinci*

THANK *you*